United Nations

Our Common Future
Chapter 10: Managing The Commons

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United Nations

1. The traditional forms of national sovereignty are increasingly challenged by the realities of ecological and economic interdependence. Nowhere is this more true than in shared ecosystems and in 'the global commons' - those parts of the planet that fall outside national jurisdictions. Here, sustainable development can be secured only through international cooperation and agreed regimes for surveillance, development, and management in the common interest. But at stake is not just the sustainable development of shared ecosystems and the commons, but of all nations whose development depends to a greater or lesser extent on their rational management.

2. By the same token, without agreed, equitable, and enforceable rules governing the rights and duties of states in respect of the global commons, the pressure of demands on finite resources will destroy their ecological integrity over time. Future generations will be impoverished, and the people who suffer most will be those who live in poor countries that can least assert their own claims in a free-for-all.

3. Management of the various commons - the oceans, outer space, and Antarctica - is at different stages of evolution, as is the very 'commonality' of these areas. In the Law of the Sea, the international community has developed one of the most ambitious and advanced of international conventions ever for the seas and the sea-bed. But a few countries have so far declined to adhere to the multilateral regime that had been the subject of protracted global negotiations, and this is blocking implementation of certain key aspects. Boundaries have been drawn on the oceans to separate the common seas from national Exclusive Economic Zones (EEZs), but as the common and claimed waters form interlocked ecological and economic systems, and as the health of one depends on the health of the other, both are discussed in this chapter. As for outer space, the least tapped global commons, discussion of joint management has only just begun. Antarctica has been covered for over a quarter of a century by a binding Treaty. Many states that are not party to it feel they should by right have a stake in the management of what they see as a part of the global commons.

I. Oceans: The Balance of Life

4. In the Earth's wheel of life, the oceans provide the balance./1 Covering over 70 per cent of the planet's surface, they play a critical role in maintaining its life-support systems, in moderating its climate, and in sustaining animals and plants, including minute, oxygen-producing phytoplankton. They provide protein, transportation, energy, employment, recreation, and other economic, social, and cultural activities.

5. The oceans also provide the ultimate sink for the by-products of human activities. Huge, closed septic tanks, they receive wastes from cities, farms, and industries via sewage outfalls, dumping from barges and ships, coastal run-off, river discharge, and even atmospheric transport. In the last few decades, the growth of the world economy, the burgeoning demand for food and fuel, and accumulating discharges of wastes have begun to press against the bountiful limits of the oceans.

6. The oceans are marked by a fundamental unity from which there is no escape, interconnected cycles of energy, climate, marine living resources, and human activities move through coastal waters, regional seas, and the closed oceans. The effects of urban, industrial, and agricultural growth are contained within no nation's Exclusive Economic Zone; they pass through currents of water and air from nation to nation, and through complex food chains from species to species, distributing the burdens of development, if not the benefits, to both rich and poor.

7. Only the high seas outside of national jurisdiction are truly 'commons'; but fish species, pollution, and other effects of economic development do not respect these legal boundaries. Sound management of the ocean commons will require management of land-based activities as well. Five zones bear on this management: inland areas, which affect the oceans mostly via rivers; coastal lands - swamps, marshes, and so on - close to the sea, where human activities can directly affect the adjacent waters; coastal waters - estuaries, lagoons, and shallow waters generally - where the effects of land-based activities are dominant; offshore waters, out roughly to the edge of the continental shelf; and the high seas, largely beyond the 200-mile EEZs of coastal states' control.

8. Major fisheries are found mostly in offshore waters, while pollution affecting them comes mostly from inland sources and is concentrated in coastal waters. Formal international management is essential in the areas

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beyond the EEZs, although greater international cooperation, including improved frameworks to coordinate national action, is needed for all areas.

1. The Balance Under Threat

9. Today, the living resources of the sea are under threat from overexploitation, pollution, and land-based development. Most major familiar fish stocks throughout the waters over the continental shelves, which provide 95 per cent of the world's fish catch, are now threatened by overfishing.

10. Other threats are more concentrated. The effects of pollution and land development are most severe in coastal waters and semi-enclosed seas along the world's shore-lines. The use of coastal areas for settlement, industry, energy facilities, and recreation will accelerate, as will the upstream manipulation of estuarine river systems through dams or diversion for agriculture and municipal water supplies. These pressures have destroyed estuarine habitats as irrevocably as direct dredging, filling, or paving. Shore-lines and their resources will suffer ever increasing damage if current, business-as-usual approaches to policy, management, and institutions continue.

> The world's environmental problems are greater than the sum of those in each country. Certainly, they can no longer be dealt with purely on a nation-state basis. The World Commission on Environment and Development must strike at this fundamental problem by recommending specific ways for countries to cooperate to surmount sovereignty, to embrace international instruments in order to deal with global threats. The growing trend towards isolationism demonstrates that the current rhythm of history is out of harmony with human aspirations, even with its chances for survival.
>
> The challenge ahead is for us to transcend the self-interests of our respective nation-states so as to embrace a broader self-interest -- the survival of the human species in a threatened world.
>
> Hon. Tom McMillan
> Minister of Environment, Government of Canada
> WCED Public Hearing
> Ottawa, 26-27 May 1986

11. Certain coastal and offshore waters are especially vulnerable to ecologically insensitive onshore development, to competitive overfishing, and to pollution. The trends are of special concern in coastal areas where pollution by domestic sewage, industrial wastes, and pesticide and fertilizer run-off may threaten not only human health but also the development of fisheries.

12. Even the high seas are beginning to show some signs of stress from the billions of tons of contaminants added each year. Sediments brought to the oceans by great rivers such as the Amazon can be traced for as much as 2,000 kilometres out to sea./2 Heavy metals from coal-burning plants and some industrial processes also reach the oceans via the atmosphere. The amount of oil spilled annually from tankers now approaches 1.5 million tons./3 The marine environment, exposed to nuclear radiation from past nuclear weapons tests, is receiving more exposure from the continuing disposal of low-level radioactive wastes.

13. New evidence of a possible rapid depletion of the ozone layer and a consequent increase in ultraviolet radiation poses a threat not only to human health but to ocean life. Some scientists believe that this radiation could kill sensitive phytoplankton and fish larvae floating near the ocean's surface, damaging ocean food chains and possibly disrupting planetary support systems./4

14. High concentrations of substances such as heavy metals, organochlorines, and petroleum have been found on the oceans' surface. With continued accumulation, these could have complex and long lasting effects./5 The sea-floor is a region of complex physical, chemical, and biological activity where microbial processes play a major role, but as yet serious damage is known to have occurred only in very localized regions. Although these findings are encouraging, given accelerating pressures and the inadequacy of present data, they provide no grounds for complacency.

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2. Oceans Management

15. Looking to the next century, the Commission is convinced that sustainable development, if not survival itself, depends on significant advances in the management of the oceans. Considerable changes will be required in our institutions and policies and more resources will have to be committed to oceans management.

16. Three imperatives lie at the heart of the oceans management question:

- The underlying unity of the oceans requires effective global management regimes.
- The shared resource characteristics of many regional seas make forms of regional management mandatory.
- The major land-based threats to the oceans require effective national actions based on international cooperation.

17. Mutual dependence has increased in recent years. The Law of the Sea Convention, with the establishment of the 200-mile EEZs, has put an additional 35 per cent of the oceans surface under national control with regard to management of natural resources. It has also provided an institutional setting that could lead to better management of these areas, given that single governments may be expected to manage more rationally resources over which they have sole control. However, this expectation ignores the realities of short sighted political and economic goals.

18. An international ecosystem approach is required for the management of these resources for sustained use. Significant gains have been made in past decades, nationally and internationally, and many essential components have been put in place. But they do not add up to a system that reflects the imperatives mentioned above. Where the EEZs of several states come together in semi-enclosed or regional seas, integrated management requires varying degrees of international cooperation, such as joint monitoring and research on migratory species and measures to combat pollution and regulate actions whose effects reach across boundaries.

19. When it comes to the high seas beyond national jurisdiction, international action is essential. The sum of the multiple conventions and programmes now in place do not and cannot represent such a regime. Even the separate UN programmes cannot easily be coordinated, given the structure of the United Nations./6

20. The Commission believes that a number of actions are urgently needed to improve regimes for oceans management. Thus the Commission proposes measures to:

- strengthen capacity for national action, especially in developing countries;
- improve fisheries management;
- reinforce cooperation in semi-enclosed and regional seas;
- strengthen control of ocean disposal of hazardous and nuclear wastes; and
- advance the Law of the Sea.

2.1 National Action

21. Coastal governments should launch an urgent review of the legal and institutional requirements for integrated management of their EEZs, and of their roles in arrangements for international cooperation. This review should be undertaken within the framework of a clear statement of national goals and priorities. Reducing overexploitation of fisheries in coastal and offshore waters might be one such goal. The rapid clean-up of municipal and industrial pollution discharging into critical marine habitats could be another. Others might include strengthening national research and management capacity, and producing an inventory of coastal and marine resources.

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22. Given the increased pressures on coastal and marine resources projected through the year 2000, all coastal states should have a complete inventory of these assets. Drawing on senior experts from national and international agencies, nations could deploy the latest satellite mapping and other techniques to put together an inventory of these resources and then monitor changes in them.

23. Many developing countries will require assistance to strengthen their legal and institutional frameworks needed for integrated management of coastal resources. Many small island and maritime developing countries lack the economic or military means to prevent the exploitation of their coastal resources or the pollution of their waters by powerful countries or companies. This has become a major concern in the Pacific in particular, and threatens the political stability of the region. International development banks and development assistance agencies should establish programmes to support the development of this Institutional capacity.

2.2 Fisheries Management

24. World fisheries have been expanding since the Second World War, with the global catch rising at a steady 6-7 per cent annually from 20 million to 65 million tons between 1950 and 1969. But after 1970, as more and more stocks were depleted, the average annual growth in catches fell to only about 1 per cent. (See Table 10-1.) With conventional management practices, the growth era in fisheries is over. Even assuming restored productivity in now depleted stocks, and an increased harvest from underutilized fisheries, FAO sees only a gradual increase in catches, perhaps rising from current levels of over 80 million tons to about 100 million. This does not augur well for future food security, especially in low-income countries where fish are a principal source of animal protein and where millions secure their livelihoods from fisheries activities./7

25. Overexploitation threatens many stocks as economic resources. Several of the world's largest fisheries - the Peruvian anchoveta, several North Atlantic herring stocks, and the Californian sardine - have collapsed following periods of heavy fishing. In some of the areas affected by these collapses, and in other rich fisheries such as the Gulf of Thailand and off West Africa, heavy fishing has been followed by marked changes in species composition./8 The reasons for these changes are not well understood, and more research is needed into the responses of marine resources to exploitation so that managers can receive better scientific advice. Greater support for such work is urgently needed, and this support must include additional assistance to developing countries in increasing their research capacity and their knowledge of their own resources.

26. One factor leading to the establishment of extended EEZs was the concern of coastal states, both industrialized and developing, over the depletion of fisheries off their coasts. A large number of conventions had been established covering most major fisheries, but they proved inadequate in most cases. Participating countries were in general unable to overcome the difficulties of allocating shares to limited common resources. Improved management was seen as an urgent need, and open access was perceived as the main obstacle to it.

27. The advent of extended EEZs under the Law of the Sea Convention was expected to solve or at least alleviate the problem. Coastal states were required to introduce effective conservation and management of the living resources in their EEZs. They could also control the activities of foreign fishermen and develop their own fisheries.

28. Industrial countries have been much more successful in doing this than developing countries. In the north-west Atlantic, the annual catch by long-range fleets has declined from over 2 million tons before 1974 to around a quarter of a million tons in 1983, and the share of the catch taken by the United States and Canada has risen from under 50 per cent to over 90 per cent.

Table 10-1

World Fish Catch in Major Fisheries, 1979-84

Region	1979	1980	1981	1982	1983	1984
			(thousand tons)			
North Atlantic	14,667	14,676	14,489	13,597	13,891	13,940
North Pacific	20,303	20,733	21,908	22,603	23,666	26,416

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Central Atlantic	6,064	6,867	6,833	7,239	7,210	1,164
Central Pacific	7,536	7,910	6,478	8,175	7,648	8,531
Indian Ocean	3,541	3,693	3,728	3,852	4,061	4,362
South Atlantic	4,420	3,895	4,037	4,340	4,314	3,957
South Pacific	7,242	6,619	7,240	8,328	6,724	8,684
Inland	7,240	7,603	8,138	8,455	9,131	9,716
Total*	71,014	71,996	74,850	76,590	76,846	82,770
Developed	37,143	38,234	38,890	39,265	37,991	42,412
Developing	33,871	33,758	35,961	37,326	36,655	40,358
Developing countries catch as per cent of world total	47.7	46.9	48.0	48.7	48.0	46.6

* Columns do not add to totals due to rounding.

Source: Based on data in FAO, Yearbooks of Fishery Statistics (Rome: 1979-84).

29. Yet long-range industrial fishing fleets still catch about 5 million tons annually in developing regions. Off West Africa, for example, over half the total catch is still taken by such fleets./9 This is due partly to the fact that many of the biggest resources lie off thinly populated areas - the western edge of the Sahara and off Namibia. But it is also due to the common lack of locally available capital, and to a shortage of local expertise in many technical aspects of fisheries, especially processing and marketing.

30. Coastal developing countries can usually obtain some modest revenue in the form of licence fees, but this represents only a fraction of what they could earn from a full national use of the resource. Another 10-15 million tons of so far underutilized or unexploited resources could be added to the existing fisheries off their coasts./10 There is a pressing need for these resources to be managed sustainably, for the benefit of developing countries and in ways that help to meet global nutritional needs.

> The opinion of the public is what you see here in this room. You see important leaders from all over Brazil, from all over the country that have come here, from the rubberman that was under a palmtree yesterday and was here speaking to the U.N. Commission and leaders that are independent. The Brazilian population yearned to have someone to speak to. Someone who will listen, who will not sort of mystify things, and someone who will not trick them. So there is an enormous expectation with regards to the seriousness of your Commission.
>
> Randau Marques
> Journalist
> WCED Public Hearing
> Sao Paulo, 28-29 Oct 1985

31. Whaling offers another example. Recognizing that the history of whaling up to the 1960s was that of overexploitation, the International Whaling Commission (IWC), the main international body regulating whaling, has taken a series of conservation measures since the early 1970s and now all stocks that are below a certain level have been classified as protected from commercial whaling.

32. In its early days, the IWC was dominated by whaling nations. After 1979, non-whaling nations became an increasingly significant majority of the membership. This change was reflected in the IWC's decisions, which increasingly opted in cases of scientific doubt for a cautious approach and the reduction of catch levels or the cessation of whaling altogether on certain stocks.

33. This trend culminated in the moratorium decision of 1982. Members have the right to object and continue commercial whaling or to catch whales for scientific purposes. There is a strongly held view in conservation circles that whaling for scientific purposes can be used as a loophole by whaling nations. Permissions for such hunting should be stringently applied by IWC members, or the IWC's credibility will be undermined.

34. An important political factor in recent developments hat been the ability of the U.S. Government to invoke legislation that enables contracts for fishing in U.S. waters to be withheld from nations that undermine marine

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conservation agreements. The value of such fishery concessions is large and the legislation has significant political and economic leverage. Another important factor has been the strength of the NGOs in organizing support for anti-whaling actions, lobbying governments and organizing boycotts of fish and other products from whaling nations.

35. By early 1987, whaling was restricted to scientific catches by Iceland and the Republic of Korea and to a small catch by Norway, which continued to object to the moratorium, but which planned to halt its commercial whaling following the 1967 season. And there were catches by Japan and the Soviet Union. The Soviet Union had indicated it would observe the moratorium after the 1987 Antarctic season, and Japan had withdrawn its objection to the moratorium with effect from 1988. However, Japan may continue whaling for scientific purposes./11 In addition, some whaling was being performed by native peoples in the Soviet Union and Alaska.

36. If the moratorium is observed and whaling for scientific purposes is not abused, commercial whaling will no longer be a major threat to the conservation of whale stocks taken as a whole. The annual rate of increase of these stocks, however, is unlikely to exceed a few per cent. Thus substantial whale populations will probably not be observed much before the second half of the next century.

2.3 Cooperation on Regional Seas

37. A large number of agreements have been entered into on regional seas. The Commission has not attempted to evaluate them all, but given the Commission's origin in the UNEP Governing Council and the General Assembly resolution, it has given special attention to UNEP's Regional Seas Programme. This programme now brings together over 130 states bordering 11 different shared seas around the world, states that have an interest in cooperating for their own and mutual benefit.

38. UNEP provides the initial impetus by bringing governments together to develop a flexible legal framework within which further agreements can be negotiated as needs require and politics allow. UNEP also provides some initial seed money for programme development, but the governments of the region themselves are meant to take over funding and management, drawing on the technical advice of UN and other agencies. The result is a gradually evolving action-oriented programme rooted in the needs of the regions as perceived by the governments concerned. Fourteen UN agencies and over 40 international and regional organizations participate in the worldwide programme.

39. The political strategy behind the programme' and the requirement that management and financing be undertaken by the participating countries have clearly been crucial to its success. But it is one thing to contribute a few million dollars for research, and quite another to incorporate the resulting findings into land-based development plans and to enforce strong pollution control programmes. The massive U.S.-Canadian clean up of the Great Lakes over the past 15 years cost $8.85 billion for partial treatment of municipal and industrial wastes./12 Huge investments will also be required to roll back land-based pollution along UNEP's regional seas. Yet nowhere have the sums been committed under agreed schedules to construct the necessary urban and industrial pollution control systems and to underwrite policies to control agricultural run-off. The programme now has to confront the regional seas challenge through the year 2000 - moving beyond general agreement on goals and research to a solid schedule of investment on a scale that will make a difference.

2.4 Measures to Control Ocean Disposal of Waste

40. The Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter (London Dumping Convention), which has world-wide application, was concluded in November 1972 and entered into force on 30 August 1975./13 Its political evolution parallels that of the International Whaling Commission. Initially, it consisted largely of dumping states, but non-dumping states are now in the majority. At present it has 61 contracting parties, and secretariat facilities are provided by the International Maritime Organization. The dumping of wastes is regulated by the three annexes to the Convention:/14 on extremely dangerous substances including high-level radioactive wastes, the dumping of which is prohibited, (Annex I); on somewhat less noxious substances, the dumping or which can be permitted only by 'prior special permit' (Annex II); and all other substances, which can be dumped only after a general permit has been obtained from national authorities (Annex III). Although the Convention applies to all wastes dumped deliberately at sea, the ocean disposal of radioactive wastes has attracted the most attention. It is this question that the Commission considers here.

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41. Prior to 1963, the United Kingdom, Switzerland, Belgium, and the Netherlands had been dumping low-level wastes regularly at the north-east Atlantic dumpsite in international waters off the coast of Spain. Despite protests from representatives of these nations at the London Dumping Convention meeting that they would ignore a moratorium resolution on low-level wastes and carry out dumping during 1983, a de facto moratorium - which all countries honour but to which some have not formally agreed - went into and remains in effect. Under it, no disposal should take place until it can be demonstrated that it is environmentally safe.

42. In 1985, the London Dumping Convention voted to extend indefinitely the moratorium on the ocean dumping of low-level radioactive wastes./15 As a result, the burden of proof that such activities are safe was effectively reversed, being put on those nations who want to dump. This revolutionary reversal, though not binding, reflects the changed composition of the London Dumping Convention.

43. In 1986, the London Dumping Convention established an intergovernmental panel of experts to examine the issue of comparative risks of land- and sea-based options for disposal of radioactive waste. Without prejudging this assessment, the Commission would urge all states to continue to refrain from disposing of either low or high- level wastes at sea or in the sea-bed. Moreover, it would seem prudent to anticipate continuing opposition to sea dumping and to actively pursue the siting and development of environmentally safe, land based methods of disposal.

> Why must we gamble with the lives of innocent children in order to generate plutonium for bombs? Even to contemplate dumping radioactive waste in waters that belong to all of us as part of our global heritage is an outrage. For us to make such important decisions on behalf of future generations without taking into account the morality of using international waters as an exclusive rubbish bin is an arrogant act.
>
> Peter Wilkinson
> Greenpeace
> WCED Public Hearing
> Oslo, 24-25 June 1985

44. Several other conventions regulate the dumping of wastes in the north-east Atlantic and North Sea, the Mediterranean Sea, and the Baltic Sea. Most of the Regional Seas Conventions also include a general provision calling on contracting parties to take all appropriate measures to prevent and reduce pollution caused by dumping.

45. Land-based sources of nuclear waste have become significant in the North Sea, where high levels of radioactivity have been found in fish, and could threaten other seas./16 The Convention for the Prevention of Marine Pollution from Land Based Sources (Paris Convention) was ratified in 1978 by eight states and the European Economic Community. While it has achieved some international cooperation, its silence on nuclear plants and its acceptance of the 'best available technology' principle in determining permitted levels of radioactive discharges clearly needs to be reviewed.

46. The Law of the Sea Convention requires states to establish national laws and regulations to 'prevent, reduce and control pollution of the marine environment from dumping'. It also requires express prior approval by the coastal state for dumping in the territorial sea, in the EEZs, and onto the continental shelf. The legislative history of this Article indicates that coastal states have not only the right to act but a duty to do so. States also have an obligation under the Law of the Sea to ensure that their activities do not injure the health and environment of neighbouring states and the commons.

47. The Commission encourages the London Dumping Convention to reaffirm the rights and responsibilities of states to control and regulate dumping within the 200-mile EEZ. It is urgent that they do so, as oceans and food chains respect no boundaries.

48. Moreover, all states should undertake to report releases of toxic and radioactive substances from land-based sources into any body of water to the appropriate Convention Secretariat so that they may begin to report on the aggregate releases into various seas. Competent authorities must be designated to keep records of the nature and quantities of wastes dumped. Beyond that, regional institutions should forward this information to the London Dumping Convention Secretariat.

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2.5 The Law of the Sea

49. The United Nations Conference on the Law of the Sea was the most ambitious attempt ever to provide an internationally agreed regime for the management of the oceans. The resulting Convention represents a major step towards an integrated management regime for the oceans. It has already encouraged national and international action to manage the oceans./17

50. The Convention reconciled widely divergent interests of states, and established the basis for a new equity in the use of the oceans and their resources. It confirmed that coastal states are empowered to exercise sovereignty over their territorial sea, sea- bed and subsoil, and the superjacent air space, up to a distance of 12 nautical miles. It redefined the rights of coastal states concerning the continental shelf. It established Exclusive Economic Zones of up to 200 nautical miles within which the coastal state may exercise sovereign rights with regard to the management of national resources, living and non-living, in the waters, sea bed, and subsoil.

51. The Convention removed 35 per cent of the oceans as a source of growing conflict between states. It stipulates that coastal states must ensure that the living resources of the EEZs are not endangered by overexploitation. Thus, governments now have not only the legal power and the self-interest to apply sound principles of resource management within this area, but they have an obligation to do so. The Convention calls for regional cooperation in formulating and implementing conservation and management strategies for living marine resources, including cooperation in the exchange of scientific information, the conservation and development of stocks, and the optimum use of highly migratory species.

52. Similarly, coastal states now have a clear interest in the sound management of the continental shelf and in the prevention of pollution from land- and sea-based activities. Under the Convention, coastal states may adopt laws and regulations for their EEZs compatible with international rules and standards to combat pollution from vessels.

53. The Convention also defines the waters, sea bed, and subsoil beyond the limits of national jurisdiction, and recognizes this as international. Over 45 per cent of the planet's surface, this sea-bed area and its resources are declared to be the 'common heritage of mankind', a concept that represents a milestone in the realm of international cooperation. The Convention would bring all mining activities in the sea-bed under the control of an International Seabed Authority.

54. By early 1987, the Convention had been signed by 159 nations, and 32 countries had ratified it. However, a small number of significant states had indicated that they were unlikely to ratify it./18 The reasons for this rest largely with the regime proposed to manage the common sea bed.

55. Despite this, many of the Convention's other provisions have been broadly accepted and have already entered into international law and practice in various ways. This process should be encouraged, especially as regards those provisions that relate to the environment. This Commission believes that the Convention should be ratified by the major technological powers and come into force. Indeed, the most significant initial action that nations can take in the interests of the oceans' threatened life-support system is to ratify the Law of the Sea Convention.

II. Space: A Key to Planetary Management

56. Outer space can play a vital role in ensuring the continued habitability of the Earth, largely through space technology to monitor the vital signs of the planet and aid humans in protecting its health. According to the 1967 Outer Space Treaty, outer space, including the moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use of occupation, or by any other means. The UN Committee on the Peaceful Uses of Outer Space has been labouring to see that these ideals remain on the agenda. This Commission, in view of these developments, considers space as a global commons and part of the common heritage of mankind.

57. The future of the space as a resource will depend not so much on technology as on the slow and difficult struggle to create sound international institutions to manage this resource. It will depend most of all upon humanity's ability to prevent an arms race in space.

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1. Remote Sensing from Space

58. If humanity is going to respond effectively to the consequences of changes human activity has induced - the build-up of atmospheric carbon dioxide, depletion of stratospheric ozone, acid precipitation, and tropical forest destruction - better data on the Earth's natural systems will be essential.

59. Today several dozen satellites contribute to the accumulation of new knowledge about the Earth's systems: for example, about the spread of volcanic gases, enabling scientists for the first time to describe the specific links between a major natural disturbance of the upper atmosphere and changes in the weather thousands of miles away./19

60. Satellites also played a key scientific role after the 1986 discovery of a 'hole' in the ozone layer over Antarctica. When ground-based observers noted this phenomenon, archived satellite data were examined and provided a record of seasonal ozone fluctuation extending back nearly a decade./20 And scientists have been able to follow closely the unfolding of the drought in the Sahel region of Africa in the 1980s. Satellite-generated maps correlating rainfall patterns and biomass have served as a tool in understanding droughts and helped in the targeting of relief aid.

> We need a kind of new earth/space monitoring system. I think that it goes farther than simply an earth environmental system. It's a combined earth/space monitoring system, a new agency that would have the resources to be able to monitor, report, and recommend in a very systematic way on the earth/space interaction that is so fundamental to a total ecological view of the biosphere.
>
> Maxwell Cohen
> University of Ottawa
> WCED Public Hearing
> Ottawa, 26-27 May 1986

61. Recently, an international and interdisciplinary group of scientists has proposed a major new initiative - the International Geosphere-Biosphere Programme (IGBP) to be coordinated through ICSU. It would investigate the biosphere using many technologies, including satellites. This proposal seemed in 1987 to be gaining momentum; it was already influencing the budget decisions of several nations on allocations for future satellite launches and is increasing coordination between existing efforts.

62. The primary frustration about this wealth of data is that the information is dispersed among governments arid institutions, rather than being pooled. UNEP's Global Environment Monitoring System is a modest effort to pool space data relevant to the Earth's habitability. It should be strengthened. But most such efforts are underfunded, undercoordinated, and inadequate to the tasks.

63. The primary responsibility for action rests initially with national governments, cooperating to pool, store and exchange data. In time, international efforts might be funded through some direct global revenue source or through contributions from individual nations. (See Chapter 12.)

2. The Geosynchronous Orbit

64. From an economic point of view, the most valuable part of the Earth's orbital space is the geosynchronous orbit, a band of space 36,000 kilometres, above the equator./21 Most communication and many weather satellites - as well as many military orbits - are in geosynchronous orbit. To prevent signals to and from the satellites interfering with one another, satellites must be placed some distance apart, effectively limiting the number that can use this valuable band to 180. Thus, the geosynchronous orbit is not only a valuable but also a scarce and limited global resource.

65. The growth in satellite communication traffic during the 1970s led to many predictions that slots would soon be saturated. Thus conflict emerged over the use and ownership of the geosynchronous orbit, largely between industrial nations that have the capacity to put satellites in this orbit and the equatorial developing nations that do not but that lie beneath this band of space.

66. The first effort to devise a property regime for geosynchronous orbit was the 1976 Bogota Declaration, signed by seven equatorial countries./22 These countries declared that the orbits above them were extensions

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of their territorial airspace. The Bogota Declaration has been challenged by some nations that see it as contradicting the 'non-appropriation' principle of the Outer Space Treaty. Another group of developing countries proposed a licensing system for the use of geosynchronous orbits./23 Countries would be awarded slots that could then be sold, rented, or reserved for future use.

67. Another way of managing this resource and capturing its rental value for the common interest would be for an international body to own and license the slots to bidders at an auction. Such an alternative would be analogous to the Seabed Authority in the Law of the Sea Convention.

66. Industrial countries have opposed the creation of a property rights regime for geosynchronous orbit, especially a regime that granted rights to slots to countries that cannot now use them. They argue that a regime of prior allocation would drive up costs and reduce the incentive of the private sector to develop and use this orbit. Others, who see a rapidly growing role for satellite communications, argue that regulatory regimes should be established before competition makes such a step more difficult.

69. Since satellite communications involve the use of radio waves, a de facto regime for the parcelling out of slots in geosynchronous orbit has emerged through the activities of the International Telecommunications Union (ITU) in the past several years. The ITU allocates the use of the radio waves (those parts of the electromagnetic spectrum used for communication)./24 The highly technical character of the task of parcelling out radio waves, combined with the fact that strict compliance is necessary to allow any user to enjoy access to this resource, has produced a successful international resource regime, based on three regional conferences, for effective management of the resource./25 Whether this approach will endure depends in large part upon the perceived justice of the decisions reached by the regional conferences.

3. The Pollution of Orbital Space

70. Debris in orbit is a growing threat to human activities in space. In 1981, a panel of experts convened by the American Institute of Aeronautics and Astronautics concluded that the growth of space debris could pose 'an unacceptable threat' to life in space within a decade./26 This debris consists of spent fuel tanks, rocket shells, satellites that no longer function, and shrapnel from explosions in space: it is concentrated in the region between 160 and 1,760 kilometres above the Earth.

71. With greater care in the design and disposal of satellites, much of it could be avoided. However, the creation of debris is an integral and unavoidable consequence of the testing and use of space weapons. The contribution of military activities to the Earth's 'debris belt' could grow greatly if plans to place large numbers of satellite based weapons and weapons-related sensors are realized.

72. The most important measure to minimize space debris, therefore, is to prevent the further testing and deployment of space based weapons or weapons designed tor use against objects in space.

73. Clean up would be expensive. It has been proposed that the major powers lead an international effort to retrieve the larger pieces of space debris from orbit. Such work would involve the design, construction, and launch of vehicles that could manoeuvre in space and grapple with large, jagged, tumbling space objects. The proposal has elicited little enthusiasm.

4. Nuclear Power in Orbit

74. Many spacecraft are nuclear-powered and threaten contamination if they fall to the Earth./27 There are two basic approaches to the problem: Ban or regulate. The option of banning all radioactive materials from space is the simplest to enact. It would eliminate the problem and would also severely stunt the further development of space-based warfare systems. A total ban should exempt scientific uses in deep-space, as small amounts of fissionable materials have been essential for the powering of deep-space probes. A ban on reactors in space would be easy to monitor, because reactors produce waste heat detectable by infrared sensors at great distances. Verifying the absence of small nuclear power systems would be more difficult, but still possible.

75. A wide variety of methods are available for regulating the use of radioactive materials in space. The most important include limiting the size of reactors permitted in orbit, requiring shielding around radioactive material sufficient to withstand reentry into the Earth's atmosphere, and requiring deep space disposal of spacecraft that

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contain radioactive material. All are technologically feasible, but would add cost and complexity to missions. Nevertheless, these measures should be implemented, as a minimum step.

5. Towards a Space Regime

76. Soon after the aeroplane was invented, it became obvious that collisions would occur unless a general air traffic control regime was established. This model offers a useful way to think about the need for and contents of a space regime. The creation of 'rules of the road' for orbital space could ensure that the activities of some do not degrade the resource for all.

> Utilization of spacecraft for solving the problems of forestry provides a good example of the peaceful use of space. Taking into account the interests of the present and future generations, there is no other more favourable area of space technology application than environmental protection, to study the natural resources of Earth and control their rational utilization and reproduction. We think that in the forthcoming years international cooperation in this field will be further expanded.
>
> L. E. Mikhailov
> USSR State Committee on Forestry
> WCED Public Hearing
> Moscow, 11 Dec 1986

77. Orbital space cannot be effectively managed by any one country acting alone. The inherently international character of orbital space has been recognized by a majority of nations in the Outer Space Treaty. The international community should seek to design and implement a space regime to ensure that space remains a peaceful environment for the benefit of all.

78. An essential step towards efficient management of the space resource is to abandon the notion that because outer space in general is unlimited, orbital space can absorb all human activity. Because of the speeds involved, orbital space is for practical purposes much 'closer' than the atmosphere. A system of space traffic control in which some activities were forbidden and others harmonized cuts a middle path between the extremes of a sole Space Authority and the present near anarchy.

79. The electromagnetic spectrum has been effectively regulated by international agreement, and through this regulation has begun to emerge the beginnings of a space regime for geosynchronous orbital space. An extension of this type of approach to control debris and the use of nuclear materials in orbit is the next logical step.

80. A fine balance must be struck between regulating activities too late and regulating non-existent activities too soon. Regulating activities on the Moon, for example, beyond the general principles laid out in the Outer Space Treaty is clearly premature. But regulating space debris and nuclear materials in Earth orbit is clearly overdue.

III. Antarctica: Towards Global Cooperation

81. The Antarctic continent - larger than the United states and Mexico combined - for over a generation has been managed under a regime of multilateral cooperation that has secured environmental protection. Signed on 1 December 1959, the Antarctic Treaty has been the vehicle for a number of important initiatives in pursuit of its two primary objectives: to maintain Antarctica for peaceful uses only, prohibiting all military activities, weapons testing, nuclear explosions, and disposal of radioactive wastes; and to promote freedom of scientific investigation in Antarctica and international cooperation to that end./28

82. The fact that the 'question of Antarctica' is today on the UN agenda/29 indicates the reality that there is a debate in the international community over the future management of the continent. Under the combined pressures of economic, technological, environmental, and other trends, there are new initiatives to establish a regime for minerals exploitation. New questions about equitable management are presenting challenges that may reshape the political context of the continent within the next decade./30

83. During the forthcoming period of change, the challenge is to ensure that Antarctica is managed in the interests of all humankind, in a manner that conserves its unique environment, preserves its value for scientific research, and retains its character as a demilitarized, non-nuclear zone of peace.

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84. Responsibility for guiding change at present rests initially with the countries party to the Antarctic Treaty./31 Eighteen nations now enjoy full decision-making status under the Treaty, with these consultative parties exercising their rights and carrying out their obligations in peaceful cooperation despite their divergent views on the territorial claims to parts of the continent. An additional 17 nations have observer status at the biennial Antarctic Treaty System (ATS) meetings.

85. The Antarctic Treaty is open to accession by any state that is a member of the United Nations, and by others invited to accede. To become a Consultative Party, a state must demonstrate concrete interest in Antarctica by conducting substantial scientific research there. The Treaty nations feel that this system is applied flexibly and opens the Treaty to all nations with a genuine interest in Antarctica. Many developing nations without the resources to conduct research on the continent feel that this condition effectively excludes most of the world's nations./32

86. But the question of participation is not polarized between industrial and developing countries. Not all industrialized countries are members of the Treaty, and Argentina, Brazil, Chile, China, India, and Uruguay have consultative status under it, while several additional developing countries have acceded to it. However, the overwhelming majority of developing countries, including all those of Africa, remain outside the arrangements.

87. There is furthermore no general agreement as to whether Antarctica is part of the international commons. For example, seven states maintain territorial claims. Moreover, many developing countries reject the idea that what they regard as the common heritage of mankind should be managed by some countries to the exclusion of others. Many of them see the Antarctic Treaty System as the exclusive preserve of the rich and technologically advanced countries. Some object to what they consider the exclusivity of the Treaty system, with countries self-appointed to determine the future of the continent. Although the Consultative Parties assert that they have managed Antarctica in the interests of all peoples, several nations maintain that these interests should not be defined by the Consultative Parties alone; this view has gained many new sources of expression since 1959. Despite the present debate over the continent's future, many nations outside the Treaty have recognized the trusteeship role played by the Treaty nations in protecting the environment of Antarctica./33

88. The Commission does not propose to adjudicate the status of Antarctica. But it sees it as essential that the continent be managed and protected in a responsible manner that takes into account the common interests at stake, it notes also that the legal and management regimes are in the midst of a process of change leading to wider participation.

89. The Antarctic Treaty Consultative Parties have endeavoured to demonstrate a strong concern for the protection of the continent's environment and the conservation of its natural resources. (See Box 10-1.) In 1964, they adopted the 'Agreed Measures for the Conservation of Antarctic Fauna and Flora',/34 which amount to a conservation protocol to the Treaty. At subsequent biennial meetings, they have continued to develop environmental principles and measures to guide the planning and execution of their activities. Additional measures would improve the scope and effectiveness of environmental protection, and it would be useful to consider means to ensure that the record of compliance with these measures is widely known.

90. The Consultative Parties have also played a leading role in the promulgation of two important international conventions relating to conservation of living resources: the 1972 Convention on the Conservation of Antarctic Seals and the 1980 Convention on the Conservation of Antarctic Marine Living Resources./35 The second arose out of concern that the depletion of Antarctic fish stocks, particularly shrimp like krill, could have severe and unpredictable effects on related and dependent species. It adopts an 'ecosystem approach' to resource management./36

91. Taken together, these legal instruments and accompanying protocols and recommendations, along with the non governmental body the Scientific Committee on Antarctic Research (SCAR), constitute what is referred to as the Antarctic Treaty System. This system demonstrates the evolution that has taken place under the Antarctic Treaty since it entered into force.

92. Several international NGOs have begun to monitor the adequacy of and compliance with environmental protection and conservation measures in Antarctica and have frequently been critical of these measures. They have also sought observer status at ATS meetings and greater involvement in the formulation and review of

Antarctic policies. Some UK agencies are concerned with southern hemisphere meteorology, oceanography, or fishing and have become involved in Antarctic science and politics. A concrete result of this interest has been invitations extended to WMO, FAO, IOC, IUCN, IWC, SCAR, and the Scientific Committee on Oceanic Research to attend as observers meetings of the Commission for the Conservation of Antarctic Marine Living Resources (CCAMLR). The European Economic Community is also a CCAMLR member as a result of its member states ceding competence to it with respect to fisheries management policies.

Box 10-1

Antarctica's Unique Treaty Arrangements

Under the Antarctic Treaty, the seven states claiming territory there have agreed with non-claimant parties to the Treaty to Bet aside the disputed territorial status of Antarctica in order to carry out agreed-upon activities in the area.

While the Treaty is in force, no acts or activities taking place will 'constitute a basis for asserting, supporting or denying a claim to territorial sovereignty in Antarctica', nor may any new claim, or enlargement of an existing claim, be asserted.

Decisions are taken by consensus, which guarantees to both claimant and non-claimant states that no activity or management practice prejudicial to their position on the territorial status of Antarctica will be approved. The Treaty provides for on-site inspection at any time in any or all areas of Antarctica by designated nationals of the consultative parties.

Source: Based on Lee Kimball, 'Testing the Great Experiment' Environment, September 1985.

93. For the ATS to remain viable into the next century, it will need to continue to evolve and adapt itself to deal with new issues and new circumstances. Although the Treaty could run indefinitely, in 1991 any of the Consultative Parties may call for a general conference of the signatory nations, to review its operation.

1. Guard Present Achievements

94. Although further change in the management status of Antarctica is inevitable, it is essential that such change not jeopardize the achievements of the Treaty System in the areas of peace, science, conservation, and environment. Antarctica has been an agreed zone of peace for nearly 30 years, free of all military activities, nuclear tests, and radioactive wastes. This is a foundation on which humanity must build.

95. Cooperation in scientific investigation has steadily expanded; it must be further strengthened, especially concerning Antarctica's role in global atmospheric and oceanic circulation and world climate. At the same time, more efforts should be made to secure full participation in such research. Means must be found to expand consultation and participation and to extend the benefits of international cooperation in Antarctic science and technology to the international community as a whole.

The most cruel environmental threat comes from the environmental movement itself as we see the animal rights laws systematically destroy our way of life and violate our right as aboriginal peoples to our traditions and values. Yet our people, including the Arctic people, need development. The challenge is to find strategies for development that meet the needs of the people and the environment.

Rhoda Inuksu
Inuit Indian
WCED Public Hearing
Ottawa, 26-27 May 1986

96. Several suggestions along these lines have been made. They include establishing a fund to facilitate the participation of interested developing countries in Antarctic science, and inviting more scientists from developing nations to join projects and visit scientific stations. Given the costly technologies involved in Antarctic science, possibilities should be explored for sharing Antarctic base and logistics capabilities with interested non-consultative states. The right to consultative status could be extended to states participating in scientific activities on a joint basis.

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97. As Antarctic activities multiply, sound conservation will also require increased data collection, monitoring, and environmental assessment. The interactive and cumulative effects of these projects must be carefully reviewed and areas of unique scientific and environmental value protected.

2. Anticipate Pressures for Mineral Development

98. Minerals of various kinds are known to exist in Antarctica, but the minerals talks have triggered false assumptions about the imminence of their development. Even given the most optimistic growth trends, it seems clear that more accessible sources will be developed elsewhere long before Antarctica attracts major investment. Only two minerals have been found that might exist in concentrations suitable for exploitation - coal in the Transantarctic Mountains and iron in the Prince Charles Mountains. Mining them would be a fool's venture./37 The costs would be prohibitive, and sufficient coal and iron can be found closer to the main markets.

99. Circumstantial evidence suggests the existence of offshore oil and gas, but no deposits have yet been discovered. The USSR, Japan, France, the United Kingdom, and the Federal Republic of Germany have surveyed Antarctica's continental shelves. The surveys were of a scientific nature, but, coinciding as they did with the first serious discussions of a minerals regime, were viewed by some observers as signalling commercial interests.

100. The 18 Consultative Parties are conducting negotiations among themselves to complete an agreed legal framework for determining the environmental acceptability of possible minerals exploration and development in Antarctica and to govern any such activities./38 Treaty members felt that it would be more difficult to agree on such a regime after actual finds have been made. The negotiations in many ways are an expression of the idea that prevention is better than cure, forethought preferable to afterthought.

101. Antarctica is an enormous continent where claims to sovereignty are in dispute and where there are no agreed legal bases for issuing licences, leasing or selling mineral rights, or receiving royalty payments. These delicate questions have now been raised and will not lie silent until they have been answered within an internationally agreed framework. Until these matters are resolved, and protection of the Antarctic environment is assured, it seems unlikely that any nation or group of nations will be able to invest securely in developing the continent's mineral resources./39

102. Given the absence of technologies tested in the ultimate extremities of Antarctic conditions, the lack of agreement on procedures to assess and take account of the impacts of any development, and the sparse data base, it could take a generation or more of dedicated research and technological development to ensure that minerals exploitation would not destroy the Antarctic's fragile ecosystem and its place in global environmental processes. Thus it is important that no minerals activity takes place until these conditions have changed, and then only in consonance with a regime that guarantees implementation of the most stringent standards needed to protect the continent's environment and share the proceeds equitably.

3. Promote Evolution of Antarctic Treaty System

103. In the years ahead, activities in Antarctica will expand in kind and scale, as will the numbers of participants in such activities. Further efforts must be made to ensure effective management of those activities and an orderly expansion of participation in such management. A variety of options are being discussed by the international community. More effective management, including expanded participation, could evolve gradually through the existing Treaty System. But given the extent of probable change and the lure of mineral wealth, however remote, such an approach could be too slow to retain political support. Another is that the above goals might be reached through the negotiation of an entirely new system. However, neither of these approaches would be free of difficulty. Yet another alternative would be to intensify efforts to make the Treaty System more universal, more open, and responsive to expressions of concrete and legitimate concern and interest in Antarctica.

4. Establish a Means for More Effective Communication

104. As activities under the different treaties increase, so does the importance of coordination among the advisory and decision making authorities responsible for various areas. Antarctica may require the

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establishment of somewhat more formal institutions than have governed the first generation of activities, in order to foster better communication and coordination both within and outside the Treaty System.

Some unique objects like Lake Baikal and Siberia, the Great Lakes in Africa and North America, are part of our global patrimony. They are some of the absolute values our planet possesses and their significance transcends any national boundaries. We should learn how to foresee their future and how to anticipate the after-effects of large-scale engineering projects.

Since people's interests vary, it cannot be taken for granted that people will accept scholars' recommendations and come to agreement on that score. And their agreement is of special importance in situations where global problems are involved and where the human race as a whole may be threatened with perils generated by the absence of such agreement.

What is needed today is the moulding of a new ethos and new arrangements for building an understanding among people, countries, and regions. And as a first step we should produce new knowledge, concentrate our research efforts on maintaining life on earth, and develop a system distributing and disseminating knowledge and new moral criteria in a way that makes it available to billions of people who inhabit our planet.

Academician N.N. Moiseev
USSR Academy of Sciences
WCED Public Hearing
Moscow, 8 Dec 1986

105. Antarctica is on the agenda of the UN General Assembly and will probably remain so. Nothing will happen, however, unless the participants in the debate find terms of reference that can command broad-based support and an agreed upon means to explore and give effect to improved management.

106. To focus on longer-term strategies to preserve and build on the achievements of the existing Treaty System, nations must create the means to foster dialogue among politicians, scientists, environmentalists, and industries from countries within and outside it. A good place to start would be the development of closer working relationships between the parties to Antarctic regimes and the international organizations within and outside the UN system that have responsibilities for science and technology, conservation, and environmental management.

107. National policy processes could also be structured to provide for dialogue with concerned industries, public interest organizations, and expert advisors, perhaps through an Antarctic advisory committee. The U.S. Government has been in the forefront of those countries appointing industry and public interest advisors to its delegations to Consultative Parties meetings. Australia, New Zealand, and Denmark have more recently followed suit.

108. Hammering out an internationally supported consensus on Antarctica is a huge task requiting time and patience. And the lure of minerals increases with every new rumour of a find. Yet such a consensus is the only way to prevent a tragic plundering of the silent continent, and to maintain Antarctica as a symbol of peaceful international cooperation and environmental protection.

Footnotes

1/ This section draws on F. Szekely, 'The Marine and Coastal Environment', prepared for WCED, 1986; J. Beddington, 'Whaling', prepared for WCED, 1986; V. Sebek, 'Policy Paper on Dumping', prepared for WCED, 1986.

2/ M.W. Holdgate et al., 'The Marine Environment', in The World Environment 1972-1982 (Dublin: Tycooly International Publishing Ltd., 1982).

3/ See National Academy of Sciences, Oil in the Sea (Washington, DC: National Academy Press, 1985); and OECD, Maritime Transport, 1984 (Paris: 1986).

4/ 'Scientists Closer to Identifying Cause of Antarctic Ozone Depletion', National Science Foundation News, 20 October 1986; Ad Hoc Working Group of Legal and Technical Experts for the Elaboration of a Protocol on the

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Control of Chlorofluorocarbons to the Vienna Convention for the Protection of the Ozone Layer (Vienna Group), 'Report of the Second Part of the Workshop on the Control of Chlorofluorocarbons, Leesburg, USA, UNEP/WG.151/Background 2, Na.86-2184, UNEP, Nairobi, 15 October 1986; A.S. Miller and I.M. Mintzer, The Sky Is the Limit: Strategies for Protecting the Ozone Layer, WRI Research Report No. 3 (Washington, DC: World Resources Institute, 1986).

5/ GESAMP in a recent evaluation of the present state of the health of the oceans, 'The Health of the Oceans', Regional Seas Reports and Studies No. 16, UNEP, Nairobi, 1982.

6/ M. Bertrand, 'Some Reflections on Reform of the United Nations', Joint Inspection Unit, United Nations, Geneva, 1985.

7/ E.P. Eckholm, Down to Earth (London: Pluto Press, Ltd., 1982).

8/ J.A. Gulland and S. Garcia, 'Observed! Patterns in Multispecies Fisheries.' in R.M. May (ed.), Exploitation of Marine Communities (Berlin: Springer-Verlag, 1984); FAO, 'Review of the State of World Fishery Resources', Fisheries Circular 710 (rev. 4), Rome, 1985.

9/ Dr J. Gulland, Marine Resources Assessment Group, Imperial College of Science and Technology, London, personal communication, 20 January 1987.

10/ FAO, op. cit.

11/ IWC, Report of the IWC 36th Session, 1986 (Cambridge: forthcoming).

12/ 1965 Report on Great Lakes Water Quality: Great Lakes Water quality Board Report to the International Joint Commission (Windsor, Ont.: IJC, 1985).

13/ IMO, 'The Provisions of the London Dumping Convention, 1972', and Decisions made by the Consultative Meetings of Contracting Parties, 1975-1984.

14/ Dumping in the Convention means any deliberate disposal at sea of material and substances of any kind, form, or description from vessels, aircraft, platform, or other artificial structures, as well as the disposal of vessels, aircraft, platforms, or other artificial structures themselves.

15/ Twenty-five nations, led by Spain, Australia, and New Zealand, supported the resolution, while Canada, France, South Africa, Switzerland, the United Kingdom, and the United States voted against.

16/ U. Grimas and A. Svansson, Swedish Report on the Skagerak (Stockholm: National Environmental Protection Board, 1985).

17/ United Nations, Final Act of the Third Conference on the Law of the Sea, Montego Bay, Jamaica, December 1982. In its final form, the Convention is composed of 17 main parts (320 articles), dealing with the territorial sea and contiguous zone; straits used for international navigation; archipelagic states; exclusive economic zone; continental shelf; high seas; regime of islands; enclosed or semi-enclosed seas; right of access of land-locked states to and from the sea and freedom of transit; the area, protection, and preservation of the marine environment; marine scientific research; development and transfer of marine technology; settlement of disputes; general provisions; and final provisions. There are nine annexes to the Convention: highly migratory species; Commission on the Limits of the Continental Shelf; basic conditions of prospecting; exploration and exploitation; statute of the International Tribunal for the Law of the Sea; Statute of the Enterprise; conciliations; arbitration and special arbitration and participation by international organizations. Under the Convention, coastal states may adopt laws and regulations in the EEZ compatible with international rules and standards to combat pollution from vessels.

18/ Among other things, declaration by the President of the United States, on 9 July 1982, and L.O.S. Bulletin, July 1985, issued by the Office of the Special Representative of the Secretary General for the Law of the Sea Convention.

19/ W. Sullivan, 'Eruption in Mexico Tied to Climate Shift Off Peru.' New York Times, 12 December 1982.

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20/ R. Kerr, 'Taking Shots at Ozone Hole Theories,' Science, 14 November 1986.

21/ When the speed of a satellite matches the speed of the planet's rotation, the satellite is stationary relative to particular places on the Earth. There is only one band or arc, directly above the equator, where it is possible to achieve geosynchronous orbit.

22/ The general case for a regulatory regime and several alternative regimes are spelled out in K.G. Gibbons, 'Orbital Saturation: The Necessity for International Regulation of Geosynchronous Orbits', California Western International Law Journal, Winter 1979

23/ A summary of Third World views is found in H.J. Levin, 'Orbit and Spectrum Resource Strategies: Third World Demands', Telecommunications Policy, June 1981.

24/ The allocation is done every 10 years at World Administrative Radio Conferences (WARCs), the last of which was held in 1979. U.S. Congress, Office oœ Technology Assessment, Radiofrequency Use and Management: Impacts from the World Administrative Radio Conference of 1979 (Washington, DC: U.S. Government Printing Office, 1980).

25/ These conferences are described in G. Coding, Jr., 'The USA and the 1985 Space WARC', and A.M. Rutkowski, 'Space WARC: The Stake of the Developing Countries, the GEO and the WARC-ORB 85 Conference', Space Policy, August 1985.

26/ AIAA Technical Committee on Space Systems, Space Debris, July 1981.

27/ The United States has launched 23 spacecraft that relied at least in part upon nuclear power sources: one source was a reactor; the rest were radioactive materials the decay heat of which is converted into electricity (thermoelectric generators). By the end of 1986 the Soviet Union had launched 31 nuclear-powered spacecraft, almost all of which contained fission reactors, and it currently operates all of the reactor-powered satellites.

28/ 'Antarctic: A Continent in Transition', Fact Sheet Folio, International Institute for Environment and Development, London, 1986.

29/ In 1983, the Seventh Summit Conference of the Non-Aligned Countries included a paragraph on Antarctica in its communique. That same year, the question of Antarctica was put on the agenda of the UN General Assembly. The debate resulted in a consensus resolution asking for the elaboration of a special report by the Secretary General, which was debated by the UN General Assembly at its 39th Session in November 1984. The consensus has not been maintained. At subsequent General Assembly sessions, resolutions on Antarctica have been passed over the objections of the parties to the Treaty, moat of whom chose not to participate in the vote.

30/ L. Kimball, 'Testing the Great Experiment', Environment, September 1985.

31/ 'Antarctic Treaty', concluded 1 December 1959 and entered into force 23 June 1961, summarized in M.J. Bowman and D.J. Harris (eds.), Multilateral Treaties: Index and Current Status (London: Butterworths, 1964).

32/ They include the original seven claimants: Argentina, Australia, Chile, France, New Zealand, Norway, and the United Kingdom; an additional five who were original signatories: Belgium, Japan, South Africa, USSR and the United States; plus six who have since acceded to the Treaty and become full Consultative Parties: Poland (1977), the Federal Republic of Germany (1981), Brazil and India (1983), and China and Uruguay (1985). Any country can accede to the Treaty, becoming a full 'Consultative Party' providing, and during such time as, it demonstrates an interest in the continent through the presence of a substantial scientific activity. Seventeen other countries have acceded to the Treaty, but do not hold consultative status. Since 1983, they have been invited to attend Antarctic Treaty meetings as observers.

33/ Both in their declaration of principles concerning the environment and in the text of the Convention on the Conservation of Antarctic Marine Living Resources, the Consultative Parties insist that the primary responsibility for these matters lies with them by virtue of their status as Consultative Parties, a proposition that Parties to the Convention who are not also Parties to the Treaty are obliged to affirm.

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34/ 'Agreed Measures for the Conservation of Antarctic Fauna and Flora', agreed 2-13 June 1984, reprinted in W.M. Bush (ed.), Antarctica and International Law (London: Oceana Publications, 1982).

35/ 'Convention for the Conservation of Antarctic Seals', concluded 11 February 1972 and entered into force 11 March 1978, summarized in Bowman and Harris, op. cit.; 'Convention on the Conservation of Antarctic Marine Living Resources', concluded 20 May 1980 and entered into force 7 April 1981, summarized in ibid. See also J.N. Barnes, 'The Emerging Convention on the Conservation of Antarctic Marine Living Resources: An Attempt to Meet the New Realities of Resource Exploitation in the Southern Ocean', in J.I. Charney (ed.), New Nationalism and the Use of Common Spaces (Totowa, NJ: Allenheld Publishers, 1982).

36/ J.A. Beddington and R.M. May, 'The Harvesting oœ Interacting Species in a Natural Ecosystem', Scientific American, November 1982.

37/ J.H. Zumborge, 'Mineral Resources and Geopolitics in Antarctica', American Scientist, January-February 1979: G. Pontecorvo, 'The Economics of the Resources of Antarctica', in Charney, op. cit.

38/ L. Kimball, 'Unfreezing International Cooperation in Antarctica', Christian Science Monitor, 1 August 1982.

39/ D. Shapley, 'Antarctic Up for Grabs', Science 82, November 1982.

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